

Agile Software Development

UNIVERSITY OF CENTRAL FLORIDA

CNT 4934

Based on “The Scrum Guide”

Required

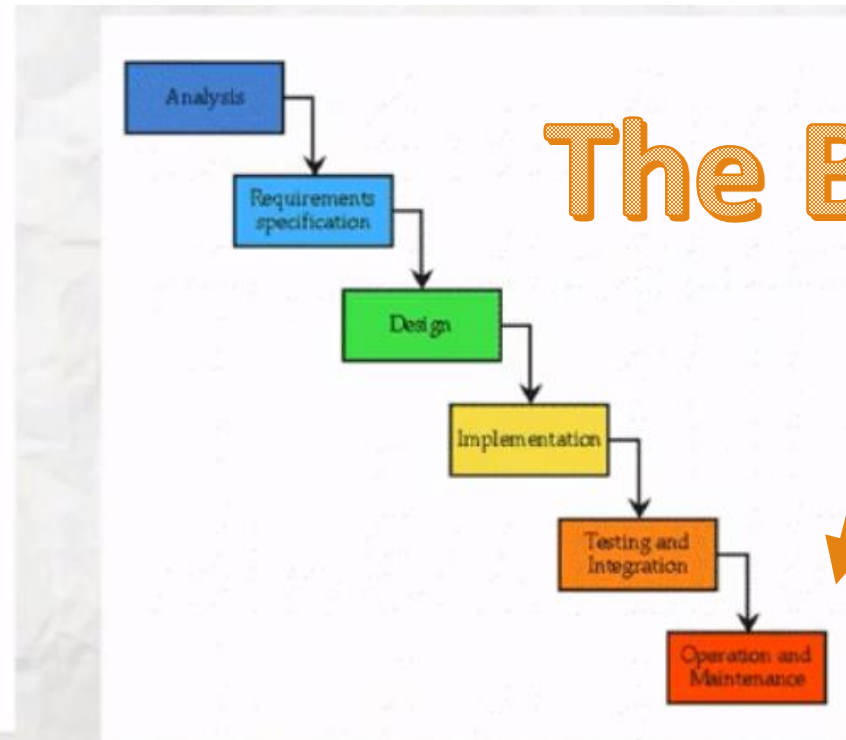
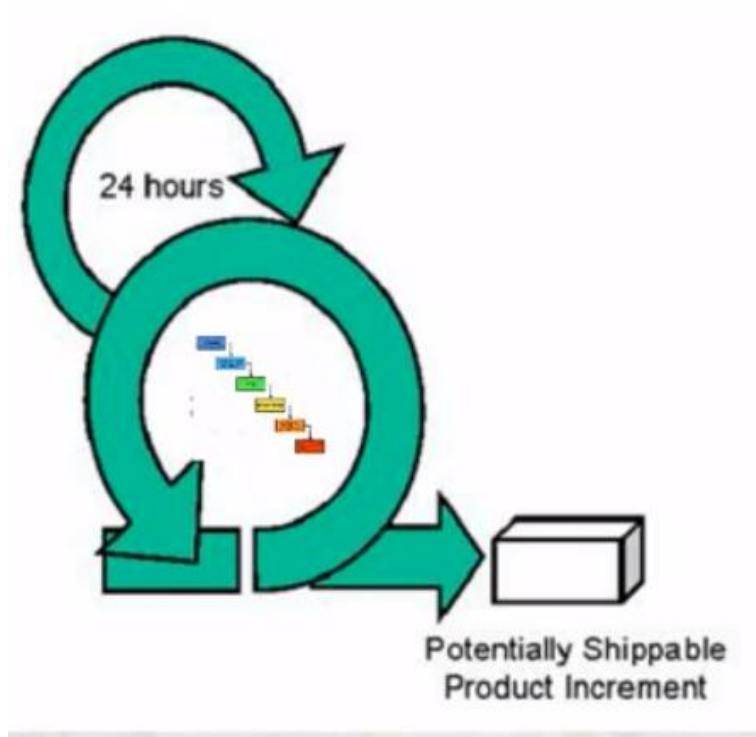
You are required to use an agile methodology.



The Need for Agile



The Birth of Agile



The Big Bang



Agile Definition

Agile is an **iterative** approach to project management and software development that helps teams deliver **value** to their customers faster and with fewer headaches.

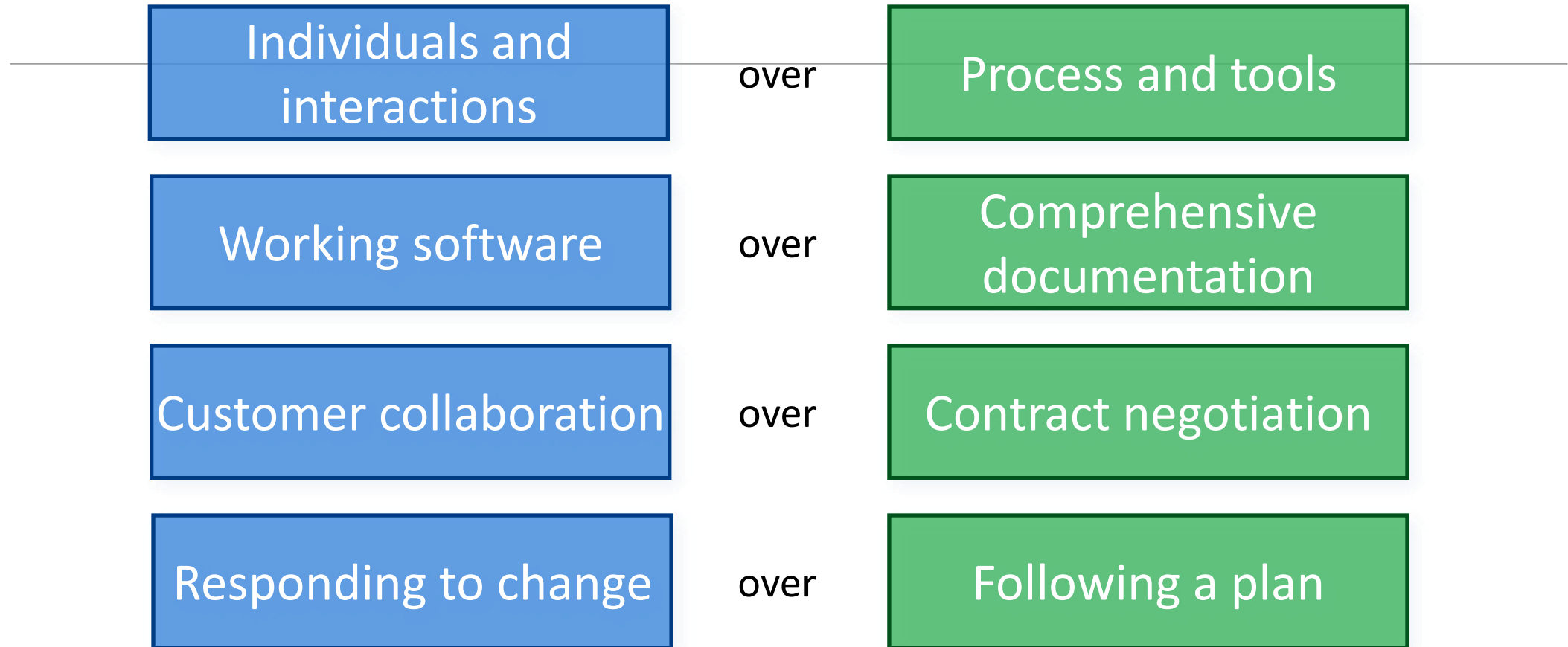
Instead of betting everything on a "big bang" launch, an agile team delivers work in small, but consumable, increments.

Key words:

Iterative

Value

Agile Manifesto

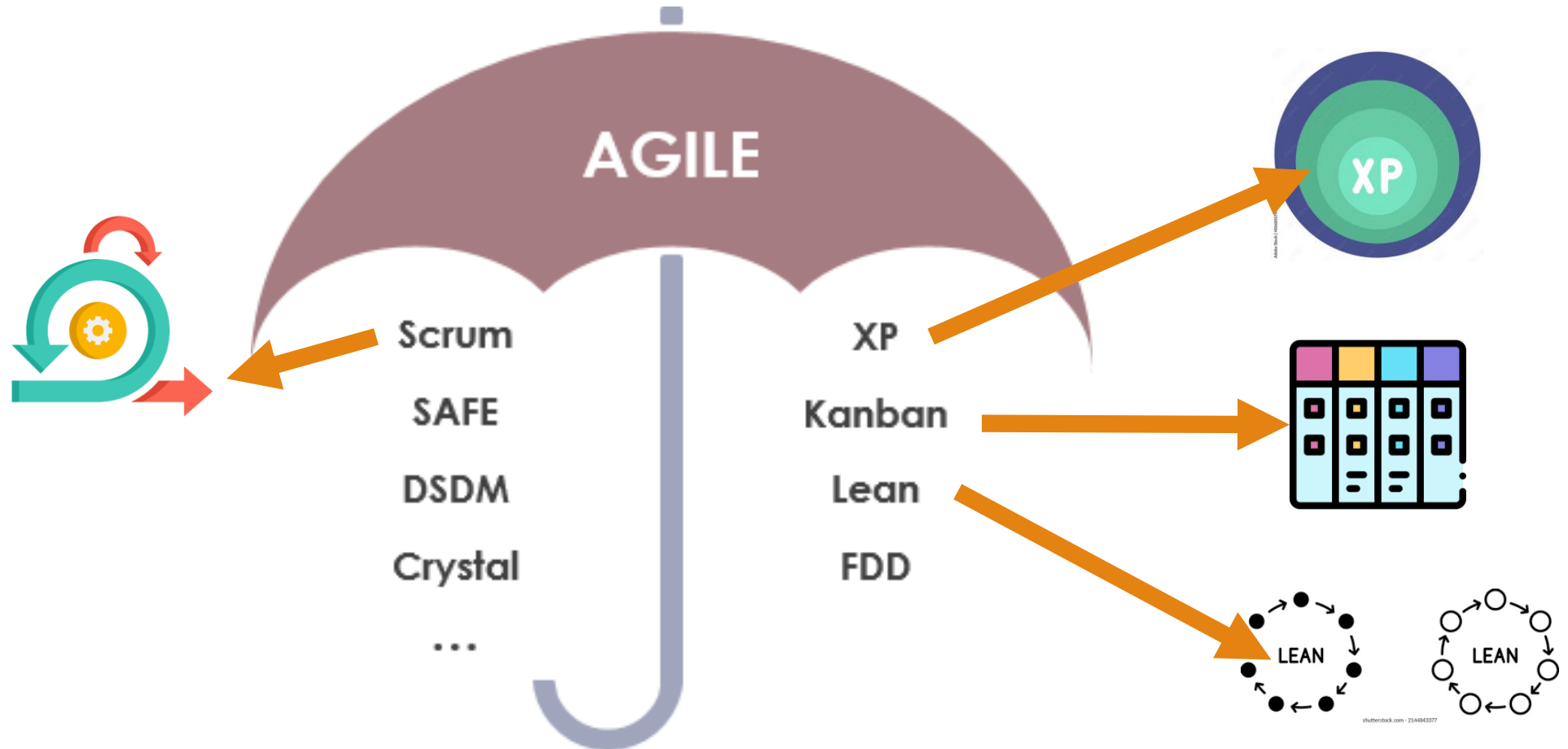


Not For Constant Ad Hoc Changes

**Constant Ad Hoc
Changes**



The Agile Umbrella



What is Scrum?

Scrum is a **lightweight framework** that helps people, teams and organizations generate value through adaptive solutions for complex problems.

In a nutshell, Scrum requires a **Scrum Master** to foster an environment where:

- A **Product Owner** orders the work for a complex problem into a Product Backlog.
- The Scrum Team turns a selection of the work into an Increment of **value** during a Sprint.
- The Scrum Team and its stakeholders inspect the results and adjust for the next Sprint.
- **Repeat**

Key Concepts:

Lightweight framework

Scrum Master fosters the correct environment



Value

More About Scrum

Scrum is simple.

Simple

Try it as is and determine if its philosophy, theory, and structure help to achieve goals and create value.

The Scrum **framework** is purposefully incomplete, only defining the parts required to implement Scrum theory.

Incomplete

Scrum is built upon by the collective intelligence of the people using it.

Rather than provide people with detailed instructions, the rules of Scrum guide their relationships and interactions.

Rules of Scrum guide

Scrum is a Wrapper

Various processes, techniques and methods can be employed within the framework.

Scrum wraps around existing practices or renders them unnecessary.

Scrum makes visible the relative efficacy of current management, environment, and work techniques, so that improvements can be made.

Leverage

Current management

Environment

Work techniques

Scrum Theory

Empiricism

Lean

Scrum is founded on **empiricism** and **lean** thinking. Empiricism asserts that knowledge comes from experience and making decisions based on what is observed. Lean thinking reduces waste and focuses on the essentials.

Scrum employs an iterative, incremental approach to optimize predictability and to control risk. Scrum engages groups of people who collectively have all the skills and expertise to do the work and share or acquire such skills as needed.

Scrum combines four formal events for inspection and adaptation within a containing event, the Sprint. These events work because they implement the empirical Scrum pillars of **transparency**, **inspection**, and **adaptation**.

Scrum Pillars

Transparency

Inspection

Adaptation

Transparency

The emergent process and work must be visible to those performing the work as well as those receiving the work.

With Scrum, important decisions are based on the perceived state of its three formal artifacts.

Artifacts that have low transparency can lead to decisions that diminish **value** and increase **risk**.



Increase risk



Diminish value

Transparency enables inspection. Inspection without transparency is misleading and wasteful.

Inspection

The Scrum artifacts and the progress toward agreed goals must be inspected frequently and diligently to detect potentially undesirable variances or problems.

To help with inspection, Scrum provides cadence in the form of its **five events**.

Inspection enables adaptation. **Inspection** —→ **Adaptation**

Inspection without adaptation is considered pointless. Scrum events are designed to provoke change.

Adaptation

If any aspects of a process deviate outside acceptable limits or if the resulting product is unacceptable, the process being applied or the materials being produced must be adjusted.

The adjustment must be made as soon as possible to minimize further deviation.

Adaptation becomes more difficult when the people involved are not empowered or self-managing.

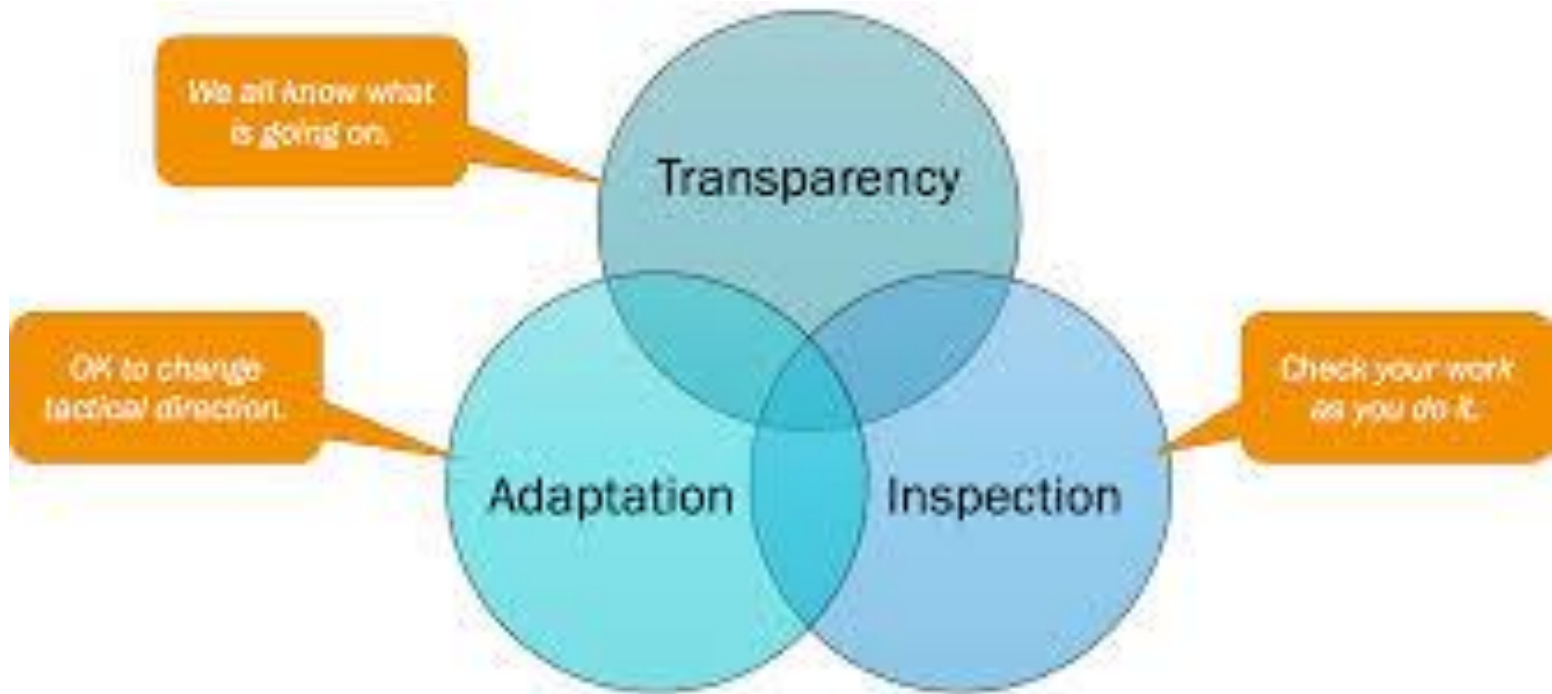
A Scrum Team is expected to adapt the moment it learns anything new through inspection.

Scrum Theory

Transparency

Inspection

Adaptation



Scrum Values



Commitment, Focus, Openness, Respect, and Courage



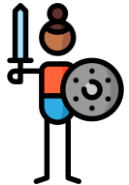
The Scrum Team **commits** to **achieving its goals** and to **supporting each other**.

Their primary **focus** is on the work of the Sprint to make the best possible **progress toward these goals**.



The Scrum Team and its stakeholders are **open about the work** and the challenges.

Scrum Team members **respect each other** to be capable, independent people, and are respected as such by the people with whom they work.



The Scrum Team members have the **courage to do the right thing**, to work on tough problems.

More About Values



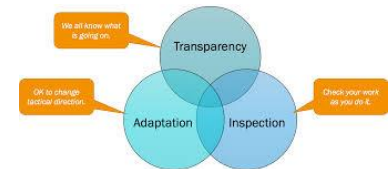
These values **give direction** to the Scrum Team with regard to their work, actions, and behavior.

The decisions that are made, the steps taken, and the way Scrum is used should reinforce these values, not diminish or undermine them.



The Scrum Team members learn and explore the values as they work with the **Scrum events and artifacts**.

When these values are embodied by the Scrum Team and the people they work with, the empirical Scrum pillars of **transparency, inspection, and adaptation** come to life building trust.



The Scrum Team

The fundamental unit of Scrum is a small team of people, a **Scrum Team**.

The Scrum Team consists of one **Scrum Master**, one **Product Owner**, and **Developers**.

Within a Scrum Team, there are no hierarchies.

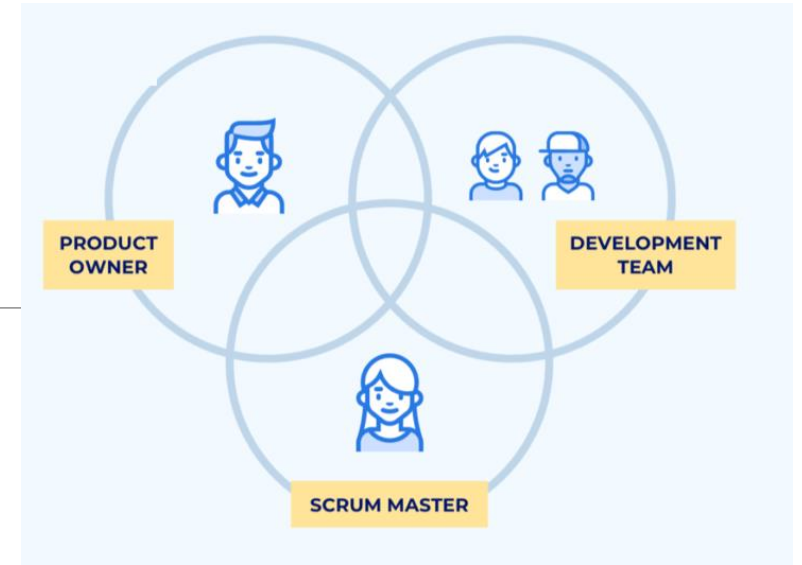
It is a cohesive unit of professionals working together towards a common objective at a time, the Product Goal.

**Cross Function =
The Team has
the skills**

Scrum Teams are **cross-functional**, meaning the members have all the skills to create value each Sprint.

Self Managing

They are also self-managing, meaning they internally decide who does what, when, and how.



Teams are Small

The Scrum Team is small enough to remain nimble and large enough to complete significant work within a Sprint, typically 10 or fewer people.

In general, smaller teams communicate better and are more productive.

If Scrum Teams become too large, they should consider reorganizing into multiple cohesive Scrum Teams, each focused on the same product.

Therefore, they should share the same Product Goal, Product Backlog, and Product Owner.

(A **Nexus Scrum** is a group Scrum teams all working on one project.)

The entire Scrum Team is accountable for creating a **valuable**, useful Increment every Sprint.

Scrum defines three specific accountabilities within the Scrum Team: the **Developers**, the **Product Owner**, and the **Scrum Master**.

Developers

Developers are the people in the Scrum Team that are committed to creating any aspect of a usable Increment each Sprint.

The specific skills needed by the Developers are often broad and will vary with the domain of work. However, the Developers are always accountable for:

Creating a plan for the Sprint, the Sprint Backlog;

- Instilling quality by adhering to a **Definition of Done**;
- Adapting their plan each day toward the Sprint Goal; and,
- Holding each other accountable as professionals.

Product Owner

Maximize Value

The Product Owner is accountable for **maximizing the value** of the product resulting from the work of the Scrum Team. How this is done may vary widely across organizations, Scrum Teams, and individuals.

The Product Owner is also accountable for effective Product Backlog management, which includes:

- Developing and explicitly communicating the Product Goal;
- Creating and clearly communicating Product Backlog items;
- Ordering Product Backlog items; and,
- Ensuring that the Product Backlog is transparent, visible and understood.

Develop the Product Goal
Create Product Backlog Items
Ordering Product Backlog Items (Prioritize)
Ensure Product Backlog Transparency

The Product Owner may do the above work or may delegate the responsibility to others. Regardless, the Product Owner remains accountable.

Delegate



Product Owner's Groove

PRODUCT OWNER

Scrum Master

Establish Scrum

The Scrum Master is accountable for establishing Scrum as defined in the Scrum Guide. They do this by helping everyone understand Scrum theory, the Scrum Team and the organization.

The Scrum Master is accountable for the Scrum Team's effectiveness, enabling the Scrum Team to improve its practices, within the Scrum framework

Scrum Masters are true leaders who serve the Scrum Team and the larger organization.

The Scrum Master serves the Scrum Team in several ways, including:

- Coaching the team members in self-management and cross-functionality;
- Helping the Scrum Team focus on creating high-value increments; **Done;**
- Causing the removal of impediments to the Scrum Team's progress; and,
- Ensuring that all Scrum events take place and are positive, productive and observe time boxes.

Coaching

Creating High-Value Increments

Helping to remove impediments

Ensuring scrum events are positive, productive and observe time boxes

More About the Scrum Master

The Scrum Master serves the Product Owner in several ways, including:

- Helping find techniques for effective Product Goal definition and management;
- Helping the Scrum Team understand the need for clear and concise Product Backlog items;
- Helping establish empirical product planning for a complex environment; and,
- Facilitating stakeholder collaboration as requested or needed.

The Scrum Master serves the organization in several ways, including:

- Leading, training, and coaching the organization in its Scrum adoption;
- Planning and advising Scrum implementations within the organization;
- Helping employees and stakeholders understand and enact an empirical approach for complex work; and,
- Removing barriers between stakeholders and Scrum Teams.

Find techniques for effective product goal definition
Help scrum team: understand need for clarity

Help planning
Facilitate stakeholder collaboration

Leading, training, and coaching
Planning and advising
Foster understanding
Removing barriers



Master of the Scrum

Scrum Events

Sprint is a container,
or wrapper

The Sprint is a container for all other events. Each event in Scrum is a formal opportunity to inspect and adapt Scrum artifacts.

designed to enable
the transparency

These events are specifically designed to enable the transparency

Failure to operate any events as prescribed results in lost opportunities to inspect and adapt.

Oh No!!!!

Events are used in Scrum to create regularity and to minimize the need for meetings not defined in Scrum.

Same Bat Time,
Same Bat Place

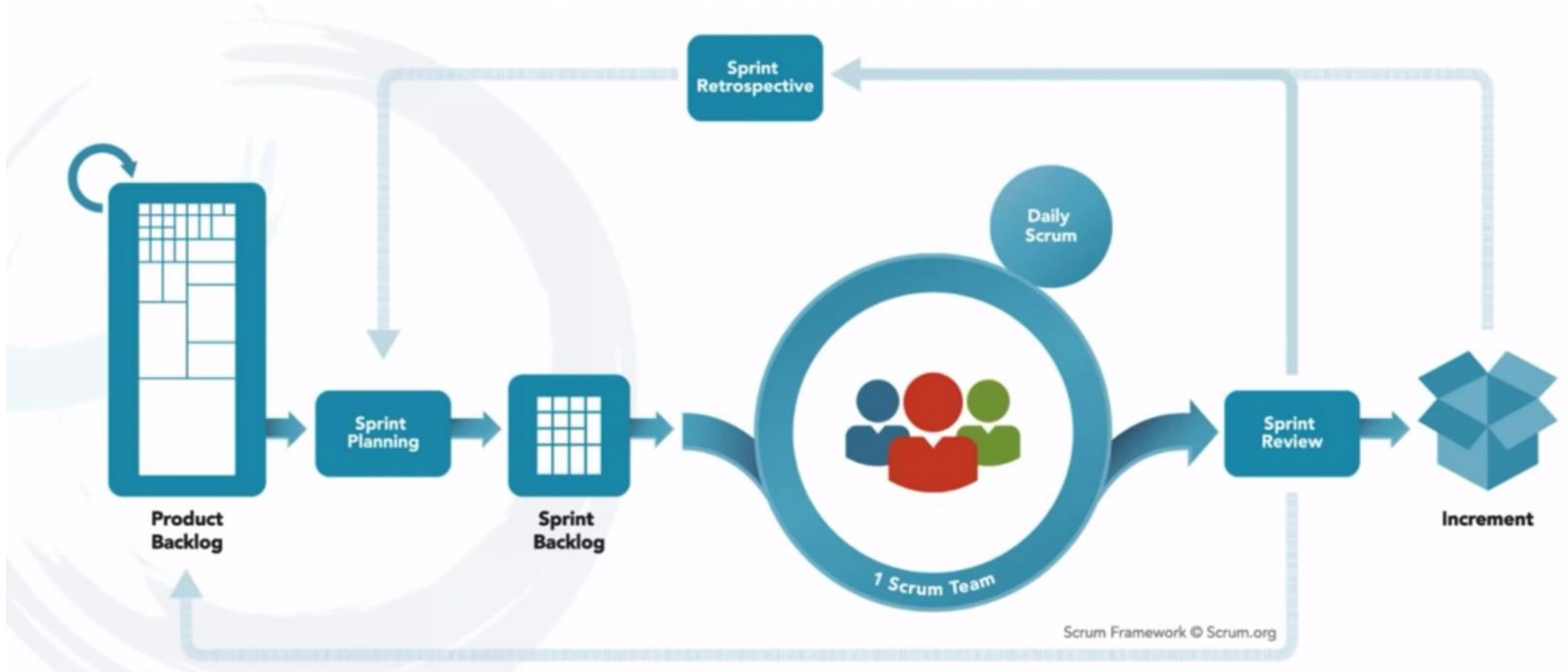
Optimally, all events are held at the **same time and place** to reduce c



FEWER MEETINGS!



Coup de grâce



The Sprint

Sprints are the heartbeat of Scrum, where ideas are turned into value.

value

They are **fixed length** events of one month or less to create consistency. A new Sprint starts immediately after the conclusion of the previous Sprint.

All the work necessary to achieve the Product Goal, including Sprint Planning, Daily Scrums, Sprint Review, and Sprint Retrospective, happen within Sprints.

During the Sprint:

- No changes are made that would endanger the Sprint Goal;
- Quality does not decrease;
- The Product Backlog is refined as needed; and,
- Scope may be clarified and renegotiated with the Product Owner as more is learned.

No endangerment of Sprint Goal
Quality does not decrease
Product backlog refined
Scope clarified

Sprints enable **predictability** by ensuring inspection and adaptation of progress toward a Product Goal at least every calendar month. When a Sprint's horizon is too long, predictability may become invalid, complexity may rise, and risk may increase. Shorter Sprints are often employed to generate more learning cycles and limit risk of cost and effort to a short time frame. Each Sprint may be considered a short project.

Predictability from:
Transparency
Inspection
Adaptation

Sprint Planning



Sprint Planning addresses the following topics:

Topic One: Why is this Sprint valuable?

The Product Owner proposes how the product could increase its **value** and utility in the current Sprint. The whole Scrum Team then collaborates to define a **Sprint Goal** that communicates why the Sprint is valuable to stakeholders. The Sprint Goal must be finalized prior to the end of Sprint Planning.

Topic Two: What can be Done this Sprint?

Through discussion with the Product Owner, the Developers select items from the Product Backlog to include in the current Sprint. The Scrum Team may refine these items during this process, which increases understanding and confidence.

Selecting how much can be completed within a Sprint may be challenging. However, the more the Developers know about their past performance (sometimes known as **velocity**), their upcoming capacity, and their Definition of Done, the more confident they will be in their Sprint forecasts.

Topic Three: How will the chosen work get done?

For each selected Product Backlog item, the Developers plan the work necessary to create an Increment that meets the Definition of Done. This is often done by decomposing Product Backlog items into smaller work items of one day or less. How this is done is at the sole discretion of the Developers. No one else tells them how to turn Product Backlog items into Increments of value.

Why is this valuable?
What can be done?
How will it get done?

Planning Poker

Planning poker, also known as “scrum poker” and “pointing poker”, is a gamified technique that development teams use to guess the effort of project management tasks.

These estimations are based on the entire group’s input and consensus, making them more engaging and accurate than other methods.

To help gauge the number of story points for the relevant tasks, teams use planning poker cards, which are similar to poker cards.



How Does It Work?

At the beginning of a poker planning session, the product owner or customer reviews an agile user story and reads it aloud.

A user story is a general and informal explanation of a software feature that describes how it will offer value to the end-user (i.e. the customer).

Step 1: Hand out the cards to participants

Distribute an identical deck of cards to everybody. Each one has a number that the team has agreed to use as their estimate. Each player should have a deck consisting of different numbers. Cohn recommended a sequence of 0, 1, 2, 3, 5, 8, 13, 20, 40, and 100.

Step 2: Read the story out loud

The moderator (either the product owner or product manager) narrates the story to the group. If participants have any questions, the moderator answers them.

More About the Game

Step 3: Discuss the story

Once the group finishes listening to the story, everyone shares their views on it. Some of these discussion points will likely include:

How should we handle the work?

How many people are expected to get involved?

What skills will be needed to work on the story?

How should we tackle any roadblocks that delay progress?

Step 4: Select and share

After the discussion, each person will privately select a card from the deck. Usually, it's used to show an estimate of story points (but can also be used to represent the number of ideal days). Once everyone selects a card, they show them at the same time.

Step 5: Reach a consensus



Daily Scrum (Standup)

The purpose of the Daily Scrum is to **inspect** progress toward the Sprint Goal and adapt the Sprint Backlog as necessary, adjusting the upcoming planned work.

The Daily Scrum is a **15-minute** event for the Developers of the Scrum Team. To reduce complexity, it is held at the **same time and place** every working day of the Sprint. If the Product Owner or Scrum Master are actively working on items in the Sprint Backlog, they participate as Developers.

The Developers can select whatever structure and techniques they want, as long as their Daily Scrum focuses on progress toward the Sprint Goal and produces an actionable plan for the next day of work. This creates focus and improves self-management.

Daily Scrums improve communications, identify impediments, promote quick decision-making, and consequently eliminate the need for other meetings.

The Daily Scrum is not the only time Developers are allowed to adjust their plan. They often meet throughout the day for more detailed discussions about adapting or re-planning the rest of the Sprint's work.

Traditional Daily Scrum Questions

What did you do yesterday to advance the sprint goal?

What are you planning to do today to advance the sprint goal?

What impediments might block your progress?

Sprint Review

The purpose of the Sprint Review is to inspect the outcome of the Sprint and determine future adaptations.

The Scrum Team presents the results of their work to key stakeholders and progress toward the Product Goal is discussed.

During the event, the Scrum Team and stakeholders review what was accomplished in the Sprint and what has changed in their environment.

Based on this information, attendees collaborate on what to do next.

The Product Backlog may also be adjusted to meet new opportunities.

The Sprint Review is a working session, and the Scrum Team should avoid limiting it to a presentation.

The Sprint Review is the second to last event of the Sprint and is timeboxed to a maximum of four hours for a one-month Sprint.

For shorter Sprints, the event is usually shorter.

Sprint Retrospective

The purpose of the Sprint Retrospective is to plan ways to increase quality and effectiveness.

The Scrum Team inspects how the last Sprint went with regards to individuals, interactions, processes, tools, and their Definition of Done. Inspected elements often vary with the domain of work.

Assumptions that led them astray are identified and their origins explored.

The Scrum Team discusses what went well during the Sprint, what problems it encountered, and how those problems were (or were not) solved.

The Scrum Team identifies the most helpful changes to improve its effectiveness.

The most impactful improvements are addressed as soon as possible.

They may even be added to the Sprint Backlog for the next Sprint.

The Sprint Retrospective concludes the Sprint.

It is timeboxed to a maximum of three hours for a one-month Sprint.

For shorter Sprints, the event is usually shorter.



Scrum Artifacts

Scrum's artifacts represent work or **value**.

They are designed to **maximize transparency** of key information.

Thus, everyone inspecting them has the same basis for **adaptation**.

Each artifact contains a commitment to ensure it provides information that enhances **transparency and focus** against which progress can be measured:

For the Product Backlog it is the **Product Goal**.

For the Sprint Backlog it is the **Sprint Goal**.

For the Increment it is the **Definition of Done**.

Product Backlog

The Product Backlog is an emergent, ordered list of what is needed to improve the product.

It is the single source of work undertaken by the Scrum Team.

Product Backlog items that can be **Done** by the Scrum Team within one Sprint are deemed ready for selection in a Sprint Planning event.

They usually acquire this degree of **transparency** after refining activities.

Product Backlog **refinement** is the act of breaking down and further defining Product Backlog items into smaller more precise items.

This is an ongoing activity to add details, such as a **description, order, and size**.

Attributes often vary with the domain of work.

The Developers who will be doing the work are responsible for the sizing.

The Product Owner may influence the Developers by helping them understand and select trade-offs.

Commitment: Product Goal

Product Backlog is a Collection of User Stories

Written in the format: As a **Type of User**, I Want to **do something**, So that **why do I want it**.

User stories are not complete without **Acceptance Criteria** and a discussion with the team to gain the full requirements of the story.

User stories cover features, functions, enhancements, and epics.

As a user, I want to reserve a hotel room so that I can spend the night in town.

As a teller, I want to be able to find clients by last name so that I can find their profile faster.

As a senior design instructor, I want to have status meetings so that I can help guide teams to success.

Collecting Requirements for Product Backlog

Establish **project goals** and objectives early.

Document every requirement activity.

Be **transparent** with requirements documentation.

Talk to the right **stakeholders** and users.

Don't make assumptions.

Confirm, confirm, confirm.

Focus on business requirements, not tools.

Prioritize your product features.

Use **UML diagrams** and block diagrams whenever possible.

Sprint Backlog

The Sprint Backlog is composed of the Sprint Goal (why), the set of Product Backlog items selected for the Sprint (what), as well as an actionable plan for delivering the Increment (how).

The Sprint Backlog is a plan by and for the Developers.

It is a highly visible, real-time picture of the work that the Developers plan to accomplish during the Sprint in order to achieve the Sprint Goal.

Consequently, the Sprint Backlog is updated throughout the Sprint as more is learned.

It should have enough detail that they can inspect their progress in the Daily Scrum.

Commitment: Sprint Goal

Increment

An Increment is a concrete stepping-stone toward the Product Goal.

Each Increment is additive to all prior Increments and thoroughly verified, ensuring that all Increments work together.

In order to provide **value**, the Increment must be usable.

Multiple Increments may be created within a Sprint.

The sum of the Increments is presented at the Sprint Review thus supporting empiricism.

However, an Increment may be delivered to stakeholders prior to the end of the Sprint.

The Sprint Review should never be considered a gate to releasing value.

Work cannot be considered part of an Increment unless it meets the **Definition of Done**.

Commitment: Definition of Done

Definition of Done

The Definition of Done is a formal description of the state of the Increment when it meets the quality measures required for the product.

Scrum Framework

Roles

- Product owner
 - Scrum Master
 - Team
-

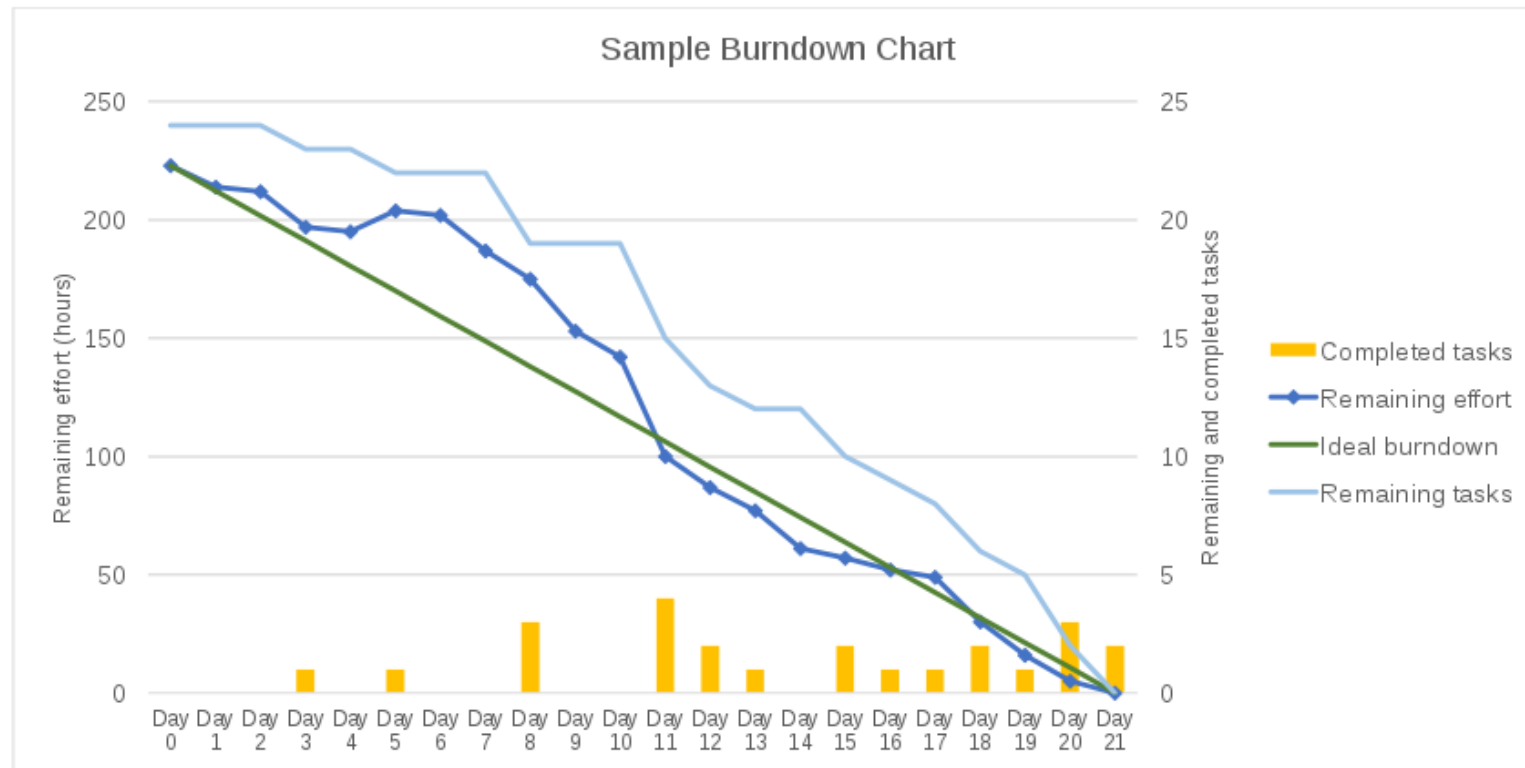
Ceremonies

- Sprint planning
- Sprint review
- Sprint retrospective
- Daily scrum meeting

Artifacts

- Product backlog
- Sprint backlog
- Increment

Burndowns



Certification

There are a number of organizations that offer certification.

Take a look at <https://www.scrumalliance.org/get-certified>

Also consider www.scrum.org



PROFESSIONAL CERTIFICATION
PROFESSIONAL SCRUM MASTER I

Rick Leinecker

has demonstrated a fundamental level of Scrum mastery, including the concepts of applying Scrum, and proven an understanding of Scrum as described in the Scrum Guide. This individual has also demonstrated a consistent use of terminology and approach to Scrum.

In recognition of this achievement, Scrum.org is pleased to award this certification.


Ken Schwaber, founder Scrum.org

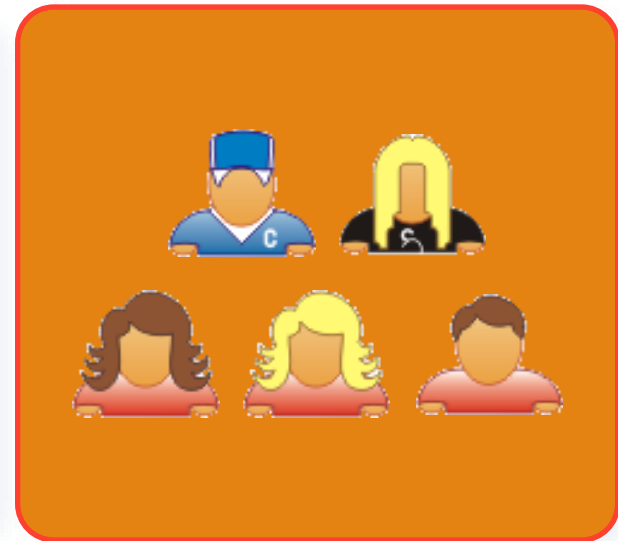
March 4, 2022

Certification Date



<https://scrum.org/certificates/768771>

Scaling: Scrum of Scrums



Scrum vs. Kanban

Kanban	Scrum
There are no pre-defined roles for a team. Although there may still be a Project Manager, the team is encouraged to collaborate and chip in when any one person becomes overwhelmed.	There are no pre-defined roles for a team. Although there may still be a Project Manager, the team is encouraged to collaborate and chip in when any one person becomes overwhelmed.
Products and processes are delivered continuously on an as-needed basis (with due dates determined by the business as needed).	Deliverables are determined by sprints, or set periods of time in which a set of work must be completed and ready for review.
Uses a “pull system,” or a systematic workflow that allows team members to only “pull” new tasks once the previous task is complete.	Also uses a “pull system” however an entire batch is pulled for each iteration.
Allows for changes to be made to a project mid-stream, allowing for iterations and continuous improvement prior to the completion of a project.	Changes during the sprint are strongly discouraged.

Jira – this is mainly intended for Agile teams

Scrum 1

Scrum Sprint 1

Implement the new weather alert system 🌤️ ⚠️ - and make over 50,000+ customers very happy 😊

🕒 35 days remaining [Complete sprint](#) ⋮

🔍 Quick filters ▾ Assignee ▾

TO DO	IN PROGRESS	DONE
SMART-43 Recalibrate the semi-coherent anomaly i... Make rocket go now   - SMART-45 Reticulate splines   - SMART-46 Align the dish   - SMART-47 SMART-17 Add app alert for changed weather event... MK-33 replicators are down.   - SMART-40 Adjust API for alert popup   - SMART-39 Update notifications settings with weather	Add app alert for changed weather events 111   4w SMART-17  SMART-3 Push notifications documentation updates Update documentation and push through channels   - SMART-29 SMART-6 Low-power indicator optimasation on mod... Draw up new schematics for power indicator panel   3d SMART-19  Reset power-indicator threshold after fatal shutdown due to low power   3h SMART-33	Recalibrate the semi-coherent anomaly in preparation to fluctuate our tachyon catalyst   - SMART-43  Hello world!   - SMART-44 SMART-17 Add app alert for changed weather event... Update documentation   - SMART-28  SMART-3 Push notifications documentation updates Review new copy   - SMART-35 Low-power indicator optimasation on model B6.2 Epic 123